

Shinhwan Kang

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Research Interest

Large Language Model, Data Augmentation, Graph Summarization, Graph Neural Network

Education

Korea Advanced Institute of Science and Technology (KAIST)

Ph.D. Candidate in Kim Jaechul Graduate School of AI (Advisor - Kijung Shin)

Seoul, Republic of Korea

Mar. 2023 - Current

Korea Advanced Institute of Science and Technology (KAIST)

M.S. Student in Kim Jaechul Graduate School of AI (Advisor - Kijung Shin)

Seoul, Republic of Korea

Mar. 2021 - Feb. 2023

Korea University

Bachelor of Engineering in Computer Science and Engineering

Seoul, Republic of Korea

Mar. 2017 - Feb. 2021

- Total GPA of 3.99/4.5 (94.2/100), Major GPA of 4.09/4.5
- Honor student (17-Spring/Fall, 19-Fall)
- Highest honor student (18-Spring/Fall, 19-Spring, 20-Spring/Fall)

Experience

ML X, NAVER

Internship

Seongnam, Republic of Korea

Mar. 2023 - Jun. 2023

- Contributed to the development of a conversational recommender system powered by HyperClove
- Explored dataset construction techniques for HyperClove fine-tuning, utilizing large language models

KAIST Data Mining Lab, KAIST

Individual Researcher (Advisor - Kijung Shin)

Daejeon, Republic of Korea

Jul. 2020 - Feb. 2021

- Participated in individual project on graph summarization algorithms

DMIS (Data Mining & Information System) Laboratory, Korea University

Undergraduate Researcher (Advisor - Jaewoo Kang)

Seoul, Republic of Korea

Mar. 2018 - Jun. 2020

- Participated in a lead optimization project and many research efforts in machine learning, especially deep learning

Brain Engineering Laboratory, Korea University

Undergraduate Researcher (Advisor - Sunhee Kim)

Seoul, Republic of Korea

Feb. 2019 - Jul. 2019

- Participated in projects on brain signal data collection and processing
- Developed real-time intelligent emotion recognition technology based on multi-modal brain signals

Publications

[C6] ReFuGe: Feature Generation for Prediction Tasks on Relational Databases with LLM Agents

WWW 2026 (ACM International World Wide Web Conference) [Paper | Github]

Kyungho Kim, Geon Lee, Juyeon Kim, Dongwon Choi, **Shinhwan Kang**, and Kijung Shin

[C5] RDB2G-Bench: A Comprehensive Benchmark for Automatic Graph Modeling of Relational Databases

NeurIPS 2025 (Advances in Neural Information Processing Systems) [Paper | Github | Document | Dataset]

Dongwon Choi, Sunwoo Kim, Juyeon Kim, Kyungho Kim, Geon Lee, **Shinhwan Kang**, and Kijung Shin

[J3] Inductive Influence Estimation and Maximization over Unseen Social Networks under Two Diffusion Models

DAMI 2025 (Data Mining and Knowledge Discovery) [Paper | Github]

Jihoon Ko*, Sojeong Kim*, Kyuhan Lee, **Shinhwan Kang**, Dongyeong Hwang, Kijung Shin, and Noseong Park

[J2] BeGin: Extensive Benchmark Scenarios and An Easy-to-use Framework for Graph Continual Learning

TIST 2025 (ACM Transactions on Intelligent Systems and Technology) [Paper | Github | Document]

Jihoon Ko*, **Shinhwan Kang***, Taehyung Kwon, Heechan Moon, and Kijung Shin

[C4] Rethinking reconstruction-based graph-level anomaly detection: limitations and a simple remedy

NeurIPS 2024 (Advances in Neural Information Processing Systems) [Paper | Github | Poster]

Sunwoo Kim, Soo Yong Lee, Fanchen Bu, **Shinhwan Kang**, Kyungho Kim, Jaemin Yoo, and Kijung Shin

[C3] Hypeboy: Generative self-supervised representation learning on hypergraphs

ICLR 2024 (International Conference on Learning Representations) [Paper | Github | Poster]

Sunwoo Kim, **Shinhwan Kang**, Fanchen Bu, Soo Yong Lee, Jaemin Yoo, and Kijung Shin

[J1] Interplay between Topology and Edge Weights in Real-World Graphs: Concepts, Patterns, and an Algorithm

DAMI 2023 (Data Mining and Knowledge Discovery) [Paper | Github | Poster]

Fanchen Bu, **Shinhwan Kang**, and Kijung Shin

[W1] Region-Conditioned Orthogonal 3D U-Net for Weather4Cast Competition

NeurIPS 2022 (Workshop Competition Track) (Advances in Neural Information Processing Systems) [Paper | Github]

Taehyeon Kim, **Shinhwan Kang**, Hyeonjeong Shin, Deukryeol Yoon, Seongha Eom, Kijung Shin, and Se-Young Yun

[C2] Personalized Graph Summarization: Formulation, Scalable Algorithms, and Applications

ICDE 2022 (International Conference on Data Engineering) (Acceptance Ratio $\approx 27\%$) [Paper | Github Appendix | Video | Slides]

Shinhwan Kang, Kyuhan Lee, and Kijung Shin

[C1] Are Edge Weights in Summary Graphs Useful? - A Comparative Study

PAKDD 2022 (Pacific-Asia Conference on Knowledge Discovery and Data Mining) (Acceptance Ratio $\approx 19.3\%$) [Paper | Github | Appendix | Slides]

Shinhwan Kang, Kyuhan Lee, and Kijung Shin

Proejects

EntireDB2AI

Software Technology Advanced Research Lab (StarLab)

- Proposed a LLM-based relational database (RDB) data augmentation method.

Republic of Korea

Mar. 2025 - Current

Deep Learning-Based Precipitation Nowcasting

National Institute of Meteorological Sciences

- Developed a classification-based precipitation nowcasting model for short-term forecasts (up to 6 hours, in 10-minute intervals)
- Processed and utilized terabyte (TB)-scale radar and satellite datasets for training and evaluation

Republic of Korea

Mar. 2021 - Feb. 2025

Extra Curricular Activities

Microsoft Student Partner Developer

Microsoft

- Recommendation system with Azure Machine Learning (Nov. 2017, Dec. 2017)
- Continuous distribution of Web App using GitHub (Jun. 2018)
- Azure Functions & Cognitive API (Jul. 2018)
- Attending the Asia-Pacific Microsoft Students Partners Summit at the National University of Singapore (NUS) in Singapore on 2018.05.18-20.

Republic of Korea

Jul. 2017 - Aug. 2018

Award/Recognition

Excellence Award Competition (3th)

Changwon Exhibition Convention Center, South
Korea

National Institute of Meteorological Sciences

Nov. 2025

- Developed a classification-based precipitation nowcasting model for short-term forecasts (up to 6 hours, in 10-minute intervals)
- Processed and utilized terabyte (TB)-scale radar and satellite datasets for training and evaluation

Weather4Cast Competition (4th)

Online

NeurIPS

Dec. 2022

- Predict super-resolution rain movies
- Transfer Learning across space and time under strong shifts
- Exploit data fusion to model graph radar and multi-band satellite images

Project for Graduation Competition (1st)

Online

Korea Univ. Computer Science and Engineering

Jun. 2020

- Lead Optimization utilizing Gene Expression Profile [\[Leader Board | Video\]](#)

AI Dataset Challenge (3rd)

Online

SELECT STAR

Apr. 2020

- Propose "Korean Blank Reasoning Dataset" Based on Crowdsourcing

Academic Services

Reviewer@SIGIR 2026

Online

ACM Special Interest Group on Information Retrieval

Reviewer@WWW 2026

Online

ACM International World Wide Web Conference

Reviewer@CIKM 2025 (Short Paper Track)

Online

ACM International Conference on Information and Knowledge Management

Reviewer@NeurIPS 2025, 2026

Online

Advances in Neural Information Processing Systems

Teaching Assistant@Upstage

Boostcamp AI Tech, Upstage

Machine Learning with Graphs

Dec. 2020 - Mar. 2021

Skills/Languages

Programming

Python, Java, C

Languages

Korean (Mother Tongue), English (Fluent)